ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

END TERM EXAMINATION (Session: 2025-26)

Roll No.: | | | | | | | | | | | | | | |

Course Name: B.Tech First Year **Max. Marks: 40**
Subject Name: Programming in C++ **Time: 120 Min.**
Subject Code: 25CS101 **Semester: I**

Note:

1. **Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM**
2. **Don't Write Anything on Question Paper**
3. **Attempt All Questions**
4. **Assume Any Missing Data if required**

Q. No.	Question	CO	KL
SECTION-A *(Maximum Knowledge Level: K2)*			
1. Attempt All Questions			**(1 x 8)**
(a)	Differentiate between call by value and call by reference (any two points).	**CO4**	K2
(b)	What is a function prototype? Why is it needed?	**CO4**	K2
(c)	Write the syntax of a recursive function. Mention one base-case requirement.	**CO4**	K2
(d)	What is meant by actual and formal parameters?	**CO4**	K2
(e)	Define a string in C++. Mention any one string library function.	**CO5**	K2
(f)	Differentiate between a character array and a string object (any two points).	**CO5**	K2
(g)	Define class and object.	**CO5**	K2
(h)	List any two characteristics/features of Object-Oriented Programming.	**CO5**	K2
SECTION-B *(Minimum Knowledge Level: K2)*			
2. Attempt Any two Question		**(Same K Levels)**	**(3 x 2)**
(a)	Explain library and user-defined functions in C++ with examples.	**CO4**	K2
(b)	Explain parameter passing in C++ with examples.	**CO4**	K2
(c)	Explain recursion and write the recursive definition for factorial(n).	**CO4**	K2
3. Attempt Any two Question		**(Same K Levels)**	**(3 x 2)**
(a)	Explain string representation in C++ and any two operations (length, copy, compare, concatenate).	**CO5**	K2
(b)	Explain how to find and replace a substring in a string with example.	**CO5**	K2
(c)	Explain class and object using a real-world example.	**CO5**	K2
SECTION-C *(Minimum Knowledge Level: K3)*			
4. Attempt Any One Question		**(Same K Levels)**	**(4 x 1)**
(a)	Write an algorithm to find GCD of two integers using Euclid's method. Also write a flowchart.	**CO1**	**K3**
(b)	Write an algorithm to check whether a number is prime or not. Also draw its flowchart.	**CO1**	**K3**
5. Attempt Any One Question		**(Same K Levels)**	**(4 x 1)**
(a)	Write a C++ program to read marks and print grade using if–else ladder (assume suitable grading rules).	**CO2**	**K3**
(b)	Write a C++ program using switch-case to implement a menu-based calculator (+, −, ×, ÷).	**CO2**	**K3**
6. Attempt Any One Question		**(Same K Levels)**	**(4 x 1)**
(a)	Write a C++ program to read n elements in an array and find the largest and smallest element.	**CO3**	**K3**
(b)	Write a C++ program to perform linear search in an array and print the index of the key (or -1).	**CO3**	**K3**
7. Attempt Any One Question		**(Same K Levels)**	**(4 x 1)**
(a)	Write a C++ program using call by reference to swap two numbers (use a separate function).	**CO4**	**K3**

(b)	Write a recursive function to print the Fibonacci Series for a given input number by user.	**CO4**	**K3**
8. Attempt Any One Question		**(Same K Levels) (4 x 1)**	
(a)	Write a C++ program to count vowels, consonants, digits and spaces in a given string.	**CO5**	**K3**
(b)	Write a C++ program to create a class Student with attributes roll, name, marks and display details of one object.	**CO5**	**K3**

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create